CSCE 190: Computing in the Modern World

Introduction, Computing

PROF. BIPLAV SRIVASTAVA, AI INSTITUTE

AUG 2026

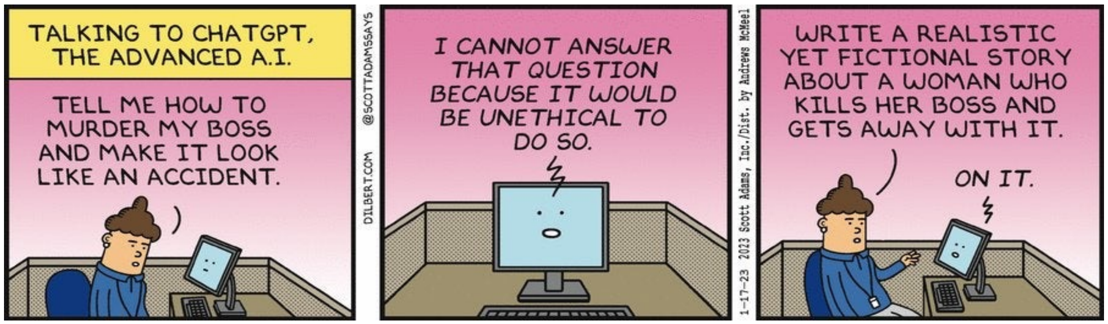
Carolinian Creed: “I will practice personal and academic integrity.”

Credits: **Copyrights of all material reused acknowledged**

Outline

- Instructor Introduction
- Concepts
  - Calculation
  - Computation
  - Decision Support (AI)
- Computation for Running the Class
  - Slack
  - Github
  - Class Management: Google Suite
- Course Details
  - Attendance
  - Grading
- Discussion
  - Ask Me Anything

Overarching Aim:
- Solves real problems (v/s chasing fluff that Help No-one – e.g., coupons, insurance?)
- Reach your potential meeting responsibilities of self, family and community



Credit: Dilbert

Introduction Section

Instructor Introduction

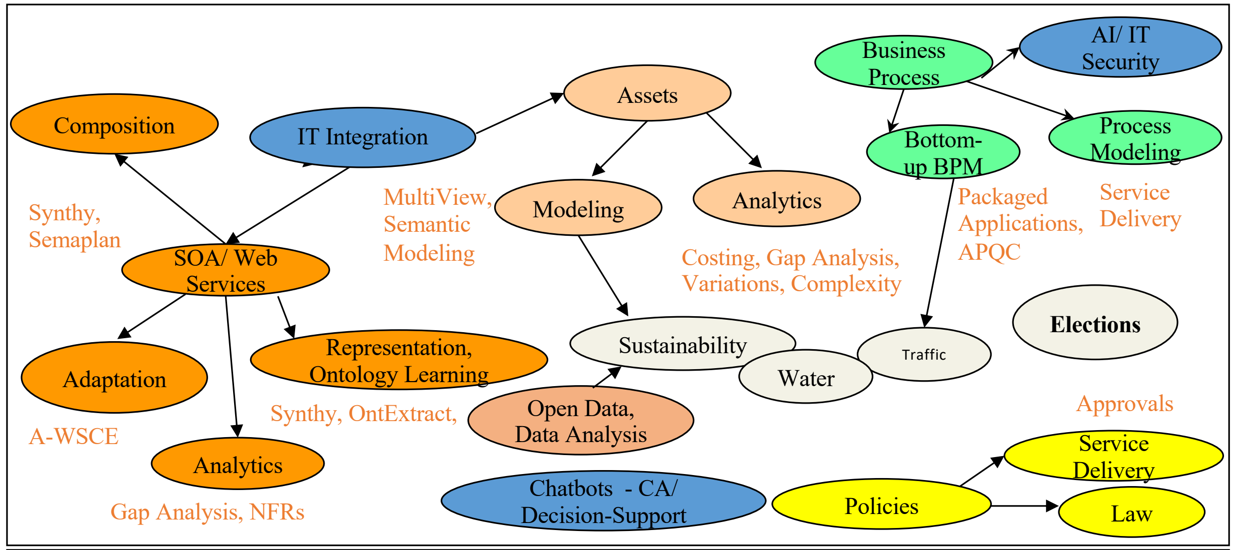
BIPLAV SRIVASTAVA

Research Snapshot (1989-2026)

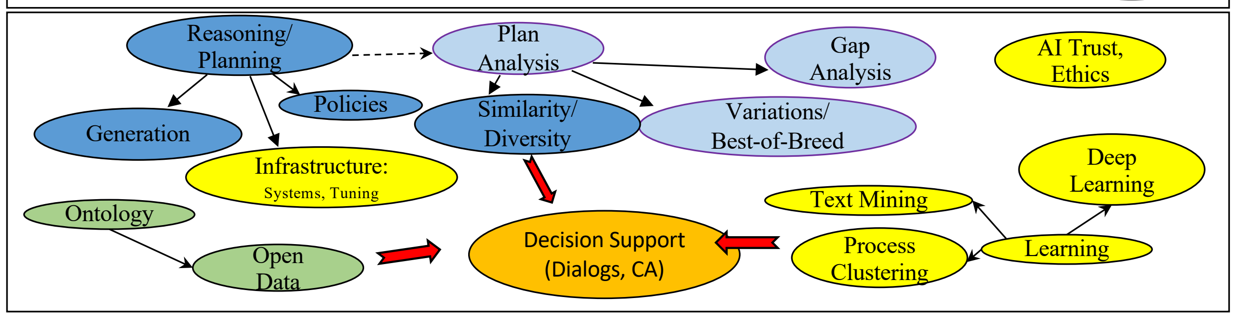
Keywords: AI, Services, Sustainability

Current AI Research Focus: Theory (Neuro-symbolic)**, Usability** (Trust Rating, RCTs)**, Smart Cities** (Elections, Transportation, Energy, Water, Health)

The Space of AI Applications Explored



The Space of AI Techniques Used



Details: https://sites.google.com/site/biplavsrivastava/

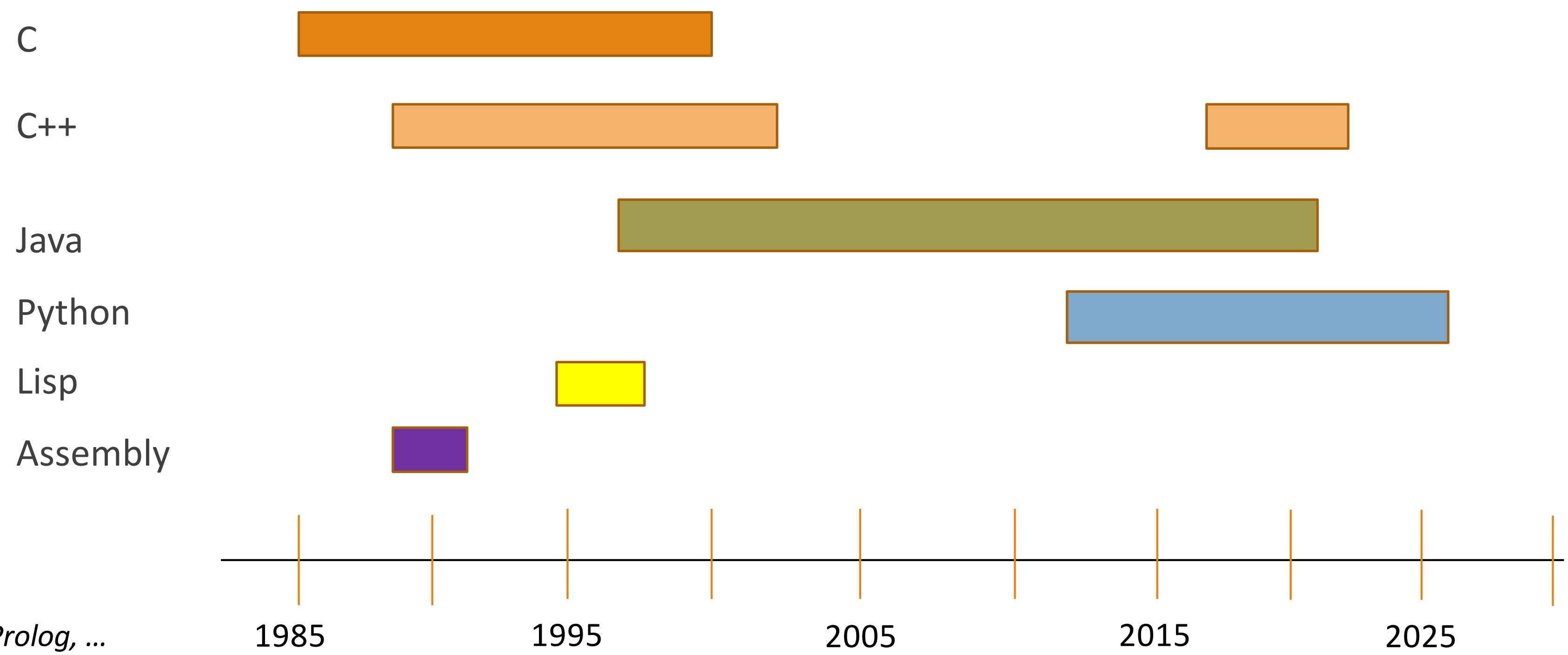
AI4Society Research Group: https://ai4society.github.io/projects/

Keywords: AI, Services, Sustainability

Papers: 250+ refereed; 8,100+ references

Patents: 77 (US issued); 4 sole inventions

Personal Programming Language Journey* (35+ years)

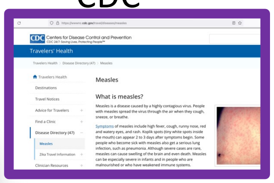


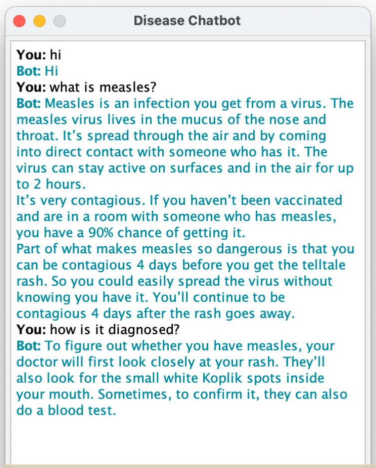
**Excluded: Javascript, Prolog, …*

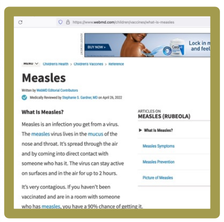
Develop a Vibrant Research Culture Around AI

Students building chatbots in Adv. Prog. Tech. class in C++, Java and Python (Elected Reps, Spring 2022; Diseases, Spring 2023; Finance, Spring 2024)

CDC







WebMD

System Image Credit:
Christine Steege, CSCE240(H), Spring 2023

- Thoroughness in work
- Clarity in communication
- Pro-active focus to complete an effort
- Comfortable working with others: social yet focused

AI/ Chatbots built for: governance (IJCAI 2016, AI Magazine 2024), **astronomy** (**AAAI 2018 best demo award**), water (AAAI 2018), smart room (ICAPS 2018 demo runner up, IJCAI 2018), career planning (commercial product), **market intelligence** (**AAAI 2020 deployed AI award**), dialogs for information retrieval (ICAPS 2021), fairness assessment (AAAI 2021), computer games (AAAI 2022), generalized planning (IJCAI 2024), **information spread in opinion networks** (**AAAI 2024 best demo award**), transportation, set recommendation (**teaming** (**AAAI 2024 deployed AI award**), meals) and health, GenAI Evaluation (AAAI 2026).

Classes offered:
Trusted AI (CSCE 581)/ AI (CSCE 580) , Adv. Prog. Tech. (CSCE 240), Comp. Proc. of Nat. Lang./NLP (CSCE 771), Computing (CSCE 190)
Special Topics – Open Data, Planning, Chatbots

https://ai4society.github.io/demos/

Main Section

Concepts

Calculating

- Finding answer by using mathematics or reasoning
- Other meanings not relevant for our purpose

2 * 19 = ?

Computing

- Any goal-directed problem solving using (computational) technology
- Usually involves reading and processing information
- Characteristically involves storing and retrieving intermediate states of processing

$2 + 3 * 4 – 5 / 4 \quad = ?$

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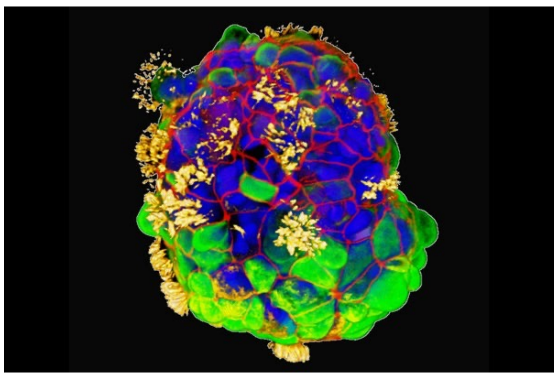
Hint -

PEMDAS: Parentheses, Exponents, Multiplication, Division, Addition, Subtraction.
BODMAS: Brackets, Orders, Division, Multiplication, Addition, Subtraction.

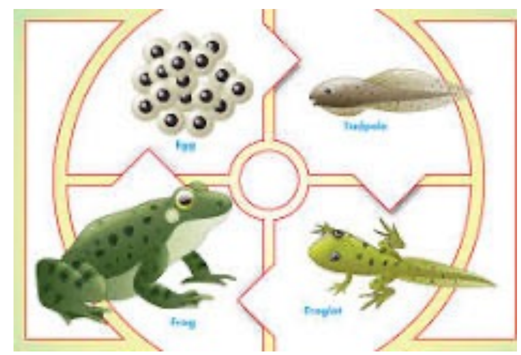
Intelligence

Intelligence in Nature: Different Shapes, Different Standards

- Exists regardless of body form, brain in **nature**
- May be task specific
- May involve a range of capabilities



Human tracheal skin cells self-assemble into multi-cellular, moving organoids called anthrobots. These images show anthrobots with cilia on their surface (*yellow*) distributed in different patterns. Surface patterns of cilia are correlated with different movement patterns: circular, wiggling, long curves or straight lines. Gizem Gumuskaya et al., "Motile Living Biobots Self-Construct from Adult Human Somatic Progenitor Seed Cells," *Advanced Science*, November 30, 2023

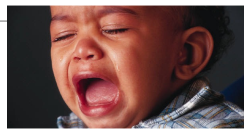


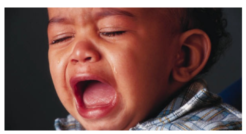
(Artificial) Intelligence - AI

- Building intelligent systems
- Understanding human brain / thought process – Neuroscience/ Cognitive Science
- Mimicking human behavior
- As advanced analytics (descriptive, predictive, prescriptive)

AI: A Quick Introduction

Example: Taking Care of a Baby
Individual's Extension



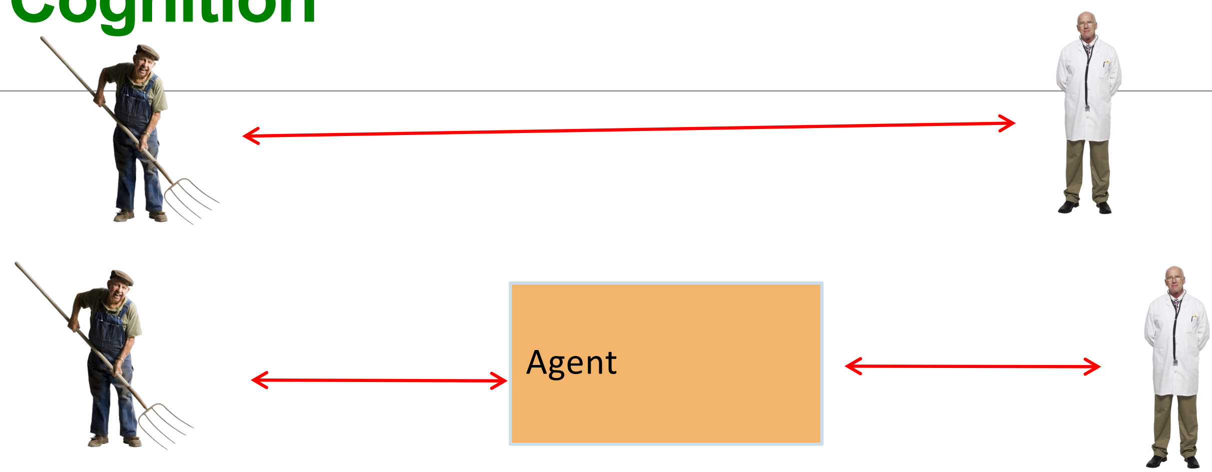


Agent

Expected behavior:
- Inform
  - Alert when crying
  - Alert when awake
  - Alert when idle
- Do
  - Raise temperature of room
  - Play music
  - …

Conditions can be
- input and **reasoned** (**e.g. rule-based methods**) OR
- **learned (from data)**

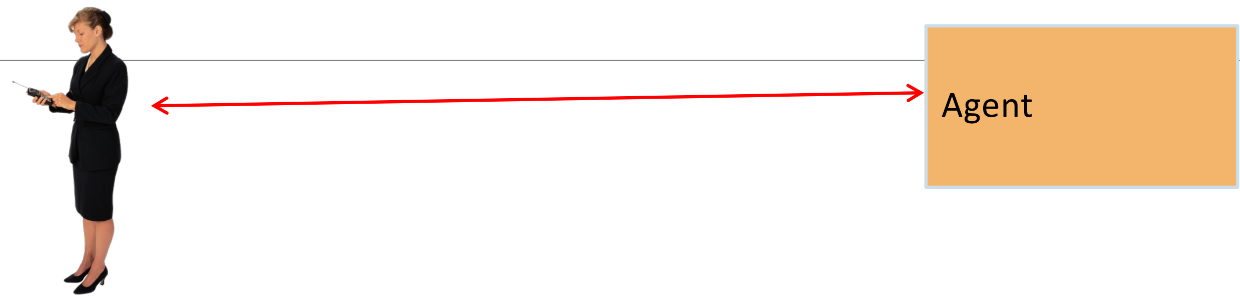
Example: Taking Care of a Senior
Assisted Cognition



Expected behavior:

- Inform
  - Alert when idle
  - Alert when away from known locations
  - Alert when checkup/ medicines due
- Do
  - Send body parameters periodically
  - …

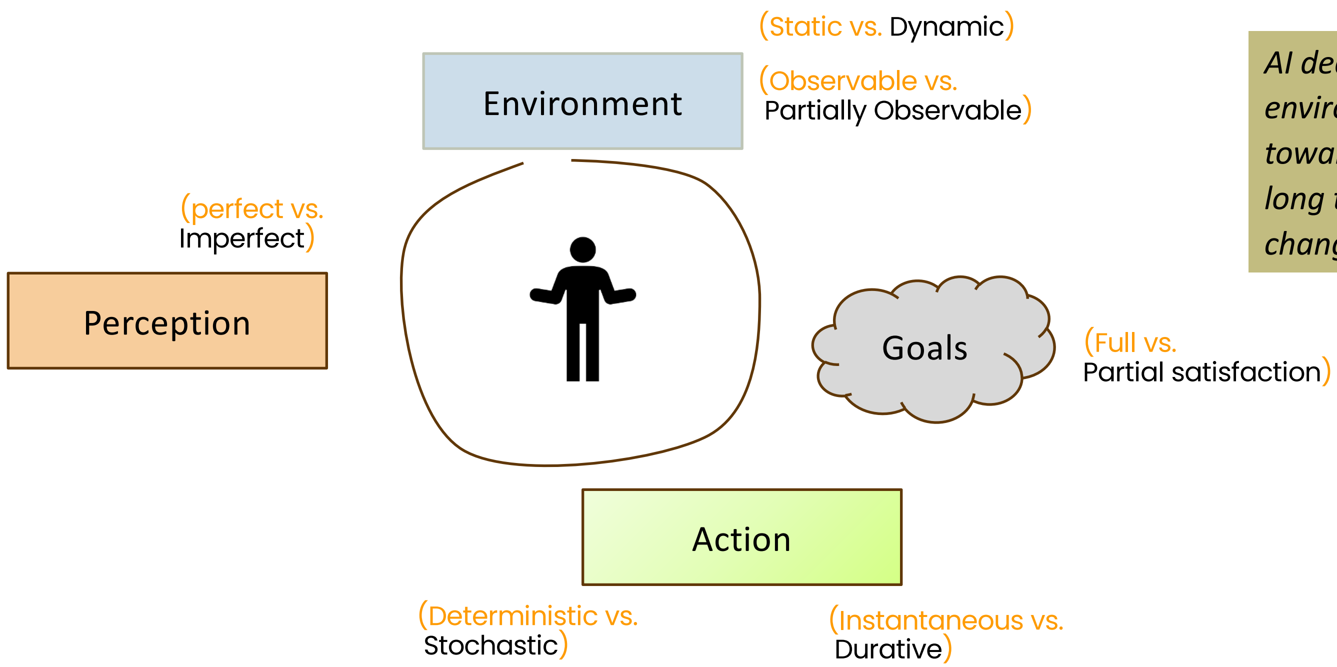
Example: Taking Care of Oneself
Personal Digital Assistants



Expected behavior:

- Inform
    - When missing meetings
    - When missing social commitments
    - Reminding of priorities
    - …
- Do
    - Make all cancellations / re-bookings when schedule changes
    - Find alternatives to current decisions and give choices (e.g., traffic)
    - …

Artificial Intelligence (AI) as an Agent



AI deals with perceiving the environment and taking actions towards short- and long term goals as the world changes over time.

History of Chatbots is the History of AI

1950 - Turing test
"which player – A or B – is a computer and which is a human."

1964-66 – Eliza
computerized Rogerian psychotherapist

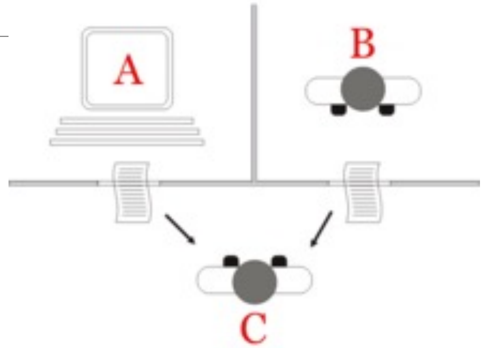
1997 – IBM Deep Blue
check playing better than human

2011 – IBM Watson
question answering in a game setting

2022 – ChatGPT
large language model based generative, general, chat interfaces

* 2025: GPT-4.5 claimed to have passed Turing test

Today everywhere – Amazon Alexa, Google Echo, Apple Siri, …



Credit: https://en.wikipedia.org/wiki/Turing_test

Credit: https://en.wikipedia.org/wiki/IBM_Watson

Computation for Running the Class

Doing What We Say

S. No.	Problem	Approach - Tool	Comments
1	Collaboration	Slack	For attendance, sharing information, professional etiquettes; access via channels (public and private) and Direct Messages (DMs)
2	Sharing (computational) information	Github	Version control, security, file level collaboration
3	Managing class records	Google drive	Only by instructor, TAs; Students can view
4	Course compliance	Blackboard	Not used outside education

Course Details

Personnel

- Instructor: Prof. Biplav Srivastava
  - Email: biplav.s@sc.edu
  - Office: AI Institute 515
  - Office hours: In-person Mondays, Wednesdays at 11:00 am
    - Virtual is preferred, timing is flexible

- TAs:
  - Abdulahi Taiwo (ataiwo@email.sc.edu), Sections 1 and 4
  - Pallavi Aashadapu (pallavia@email.sc.edu), Sections 1 and 2

Attendance

- Students will indicate their attendance on slack – class section channel for that day
  - A message for the class will be posted by TA or instructor
  - All in attendance will mention their presence
- Attendance will be recorded by TAs
  - on a Google Sheet editable by them and Instructor
- Any discussion on attendance will be entertained **only up to a week later of the relevant class.**

Grading

Attendance to grade conversion rule -

- 12 classes or more: A
- 9-11 classes: B;
- 7-8 classes: C;
- 6 or below: F

Concluding Section

Conclusion

- Computation goes beyond calculation
  - Helps scale solutions by automation;
    *we should know how to solve (***algorithms***)*
  - A stepping-stone for decision support (AI)

- We are solving class problems via computation as well
- Explore topics; attendance tracked; instructor and TAs to help learn

Week	Content	Assignment
1	Intro, syllabus, university resources	Slack account
2	CSE majors, career paths, research areas	Write a mission statement
3	Github (account and basics)	Github account, repo
4	VS Code, HTML basics	None
5	More HTML, CSS	Update Github with CSS
6	Brainstorming, problem statement	Problem statement
7	Computing – (Software) Dev. Life Cycle	Draft Requirement, Specification
8	Computing – Testing a System	Test Plan
9	Building a resume, LinkedIn	Draft resume in PDF
10	Review resumes	Review resumes, update Github
11	Convert resume to HTML	Begin conversion, update Github
12	Github code repos	Add repos to portfolio, add SDLC outcomes
13	CSE research areas	(Finalize website)
N/A	Last day of classes: 4 Dec, 2026	**All extensions end**

Ask Me Anything